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Optical module with two differential transmission line pairs with different lengths designed to cancel phase shift and stabilize high-frequency signal

Abstract

Each of the pair of lead pins has a first portion protruding from the first surface and a second portion protruding from the second surface. Each of the pair of first lines is connected to the first portion. Each of the pair of second lines is connected to the second portion. A photoelectric device is electrically connected to the pair of first lines. A longer one of the pair of first lines is electrically connected to a shorter one of the pair of second lines, through a corresponding one of the pair of lead pins. A shorter one of the pair of first lines is electrically connected to a longer one of the pair of second lines, through another corresponding one of the pair of lead pins.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims priority from Japanese patent application JP2020-083403 filed on May 11, 2020, the contents of which are hereby incorporated by reference into this application.

BACKGROUND

1. Field

(2) The disclosure relates to an optical module.

2. Description of the Related Art

(3) Optical modules are known which convert between electrical and optical signals.

WO2018/219318A1 discloses a structure that enables stable transmission of a high-frequency signal in a single-ended transmission line. JP2005-191088A discloses a structure to which a differential transmission line pair is applied.

(4) The differential transmission line pair can stabilize the high-frequency signal by transmitting signals with phases inverted by 180 degrees, through a pair of lines. However, difference in length of the pair of lines results in phase shift, making the high-frequency signal unstable.

SUMMARY

(5) The disclosure aims at stabilizing a high-frequency signal.

(6) (1) An optical module includes: a conductive block having a first surface and a second surface; a pair of lead pins passing through and insulated from the conductive block, each of the pair of lead pins having a first portion protruding from the first surface and a second portion protruding from the second surface; a first substrate having a pair of first lines with different lengths, each of the pair of first lines connected to the first portion, the pair of first lines being a first differential transmission line pair; a second substrate having a pair of second lines with different lengths, each of the pair of second lines connected to the second portion, the pair of second lines being a second

differential transmission line pair; and a photoelectric device electrically connected to the pair of first lines, the photoelectric device configured to convert an optical signal and an electrical signal at least from one to another. A longer one of the pair of first lines is electrically connected to a shorter one of the pair of second lines, through a corresponding one of the pair of lead pins. A shorter one of the pair of first lines is electrically connected to a longer one of the pair of second lines, through another corresponding one of the pair of lead pins.

(7) The longer first line and the shorter second line are electrically connected, and the shorter first line and the longer second line are electrically connected, thereby canceling phase shift and stabilizing a high-frequency signal.

(8) (2) In the optical module according to (1), the first substrate may have a surface, on which the pair of first lines are formed, expanding in a direction intersecting the first surface. The second substrate may have a surface, on which the pair of second lines are formed, expanding along the second surface.

(9) (3) In the optical module according to (2), the pair of lead pins may extend in a first direction and are arranged adjacent to each other in a second direction intersecting the first direction. The pair of first lines may be arranged adjacent to each other in the second direction. The pair of second lines may extend from the pair of lead pins in a third direction intersecting both the first direction and the second direction, curve, and extend in one of both ways along the second direction.

(10) (4) In the optical module according to (3), the one of both ways may be a direction from the shorter one to the longer one of the pair of first lines.

(11) (5) In the optical module according to any one of (1) to (4), the pair of first lines may have respective first curved portions bent in a same direction, one of the respective first curved portions on an outer side being longer. The pair of second lines may have respective second curved portions bent in a same direction, one of the respective second curved portions on an outer side being longer.

(12) (6) In the optical module according to (5), the pair of second lines may have the respective second curved portions adjacent to respective junctions with the pair of lead pins.

(13) (7) In the optical module according to (6), the pair of second lines may have the respective second curved portions at positions overlapping with the first substrate in a direction in which the pair of lead pins extend.

(14) (8) The optical module according to any one of (1) to (7), may further include a mounting substrate on which the photoelectric device is mounted. The first substrate may be located deviating in a direction from a center of the first surface to an end portion thereof. The mounting substrate may be located deviating in a direction closer to the first substrate from the center of the first surface.

(15) (9) The optical module according to (8), may further include: a thermoelectric cooler having an upper surface and a lower surface, the thermoelectric cooler at the lower surface fixed to the center of the first surface of the conductive block, the thermoelectric cooler containing a Peltier device configured to transfer heat between the upper surface and the lower surface; and a support block fixed to the upper surface of the thermoelectric cooler, the support block located deviating in a direction closer to the first substrate from the center of the first surface. The mounting substrate may be mounted on the support block.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a side view of an optical module in an embodiment.

(2) FIG. 2 is a perspective view of a conductive block and electronic components mounted on it.

(3) FIG. 3 is a perspective view, opposite to FIG. 2, of the conductive block and the electronic

components mounted on it.

(4) FIG. 4 is a plan view of a pair of lead pins and a second substrate.

(5) FIG. 5 is a diagram of frequency characteristics, in a reference example, obtained by simulation using a three-dimensional electric field analysis tool.

(6) FIG. 6 is a diagram of frequency characteristics, in the embodiment, obtained by simulation using the three-dimensional electric field analysis tool.

DETAILED DESCRIPTION

(7) Hereinafter, some embodiments will be described specifically and in detail with reference to drawings. In all the drawings, the members with the same reference numerals have the identical or same feature and their repetitive description will be omitted. Sizes of figures do not always comply with magnification.

(8) FIG. 1 is a side view of an optical module in an embodiment. The optical module **100** is a TO-CAN (Transistor Outline-Can) type optical module, and may be any of a transmitter optical sub-assembly (TOSA) with a light emitting device, a receiver optical sub-assembly (ROSA) with a light receiving device, and a bi-directional optical sub-assembly with both a light emitting device and a light receiving device (BOSA). The optical module **100** has a conductive block **10** (e.g., eyelet).

(9) FIG. 2 is a perspective view of the conductive block and electronic components mounted on it. FIG. 3 is a perspective view, opposite to FIG. 2, of the conductive block **10** and the electronic components mounted on it.

(10) The conductive block **10** is made of a conductive material such as metal, and has a first surface **12** and a second surface **14**. The conductive block **10** has some through holes **16** penetrating between the first surface **12** and the second surface **14**. The conductive block **10** has a pedestal portion **18** integral to the first surface **12**. The pedestal portion **18** is raised on the first surface **12**. The pedestal portion **18** is also made of a conductor. The conductive block **10** is connected to a reference potential, e.g., ground.

(11) [Pair of Lead Pins]

(12) The optical module **100** includes a pair of lead pins **20A**, **20B**. The pair of lead pins **20A**, **20B** extend in a first direction **D1**. The pair of lead pins **20A**, **20B** are arranged adjacent to each other in a second direction **D2** intersecting or perpendicular to the first direction **D1**. Each of the pair of lead pins **20A**, **20B** passes through and is insulated from the conductive block **10**. For example, the through hole **16** of the conductive block **10** is filled with an insulating material **22** such as glass. Each of the pair of lead pins **20A**, **20B** has a first portion **24** protruding from the first surface **12**. Each of the pair of lead pins **20A**, **20B** has a second portion **26** protruding from the second surface **14**.

(13) [First Substrate]

(14) The optical module **100** has a first substrate **28** (e.g., a relay substrate). The first substrate **28** is mounted on the pedestal **18**. The first substrate **28** has a pair of first lines **30S**, **30L** formed thereon, constituting a first differential transmission line pair. Each of the pair of first lines **30S**, **30L** is bonded to the first portion **24** of a corresponding one of the pair of lead pins **20A**, **20B**. Instead of using a wire, a filler metal **32** (solder, brazing material) is used for the bonding.

(15) The first substrate **28** has a surface, on which the pair of first lines **30S**, **30L** are formed, that spreads in a direction (e.g., the first direction **D1**) intersecting (e.g., perpendicular to) the first surface **12**. The first substrate **28** deviates in a direction from the center of the first surface **12** to the end portion. The pair of first lines **30S**, **30L** are arranged adjacent to each other in the second direction **D2**. The pair of first lines **30S**, **30L** have first curved portions **34S**, **34L**, respectively. The first curved portions **34S**, **34L** of the pair of first lines **30S**, **30L** are bent in the same direction. The first curved portion **34L** on the outer side is longer, whereby the pair of first lines **30S**, **30L** are different in length from each other.

(16) [Second Substrate]

(17) The optical module **100** includes a second substrate **36** such as a flexible substrate (FPC). The second substrate **36** is opposed to the second surface **14**. As shown in FIG. 1, a printed circuit board (PCB) **102** is connected to the second substrate **36**.

(18) The second substrate **36** has a pair of second lines **38S**, **38L** formed thereon, constituting a second differential transmission line pair. Each of the pair of second lines **38S**, **38L** is bonded to the second portion **26** of a corresponding one of the pair of lead pins **20A**, **20B**. The second substrate **36** has a surface, on which the pair of second lines **38S**, **38L** are formed, that spreads along the second surface **14**.

(19) FIG. 4 is a plan view of the pair of lead pins **20A**, **20B** and the second substrate **36**. Each of the pair of second lines **38S**, **38L** extends from a corresponding one of the pair of lead pins **20B**, **20A**. The lead-out direction is a third direction **D3** that intersects (e.g., perpendicular to) both the first direction **D1** and the second direction **D2**.

(20) The pair of second lines **38S**, **38L** are bent and have second curved portions **40S**, **40L**, respectively. The pair of second lines **38S**, **38L** have second curved portion **40S**, **40L** adjacent to junctions with the lead pins **20B**, **20A**, respectively. The second curved portions **40S**, **40L** overlap with the first substrate **28** in a direction (first direction **D1**) in which the pair of lead pins **20B**, **20A** extend (FIG. 2).

(21) The second curved portions **40S**, **40L** of the pair of second lines **38S**, **38L** are bent in the same direction. The second curved portion **40L** on the outer side is longer, whereby the pair of second lines **38S**, **38L** are different in length from each other. The pair of second lines **38S**, **38L** extend in one of both ways along the second direction **D2** from the second curved portion **40S**, **40L**, respectively. The one of both ways is, for example, the direction from the shorter first line **30S** (corresponding one lead pin **20A**) to the longer first line **30L** (corresponding other lead pin **20B**).

(22) [Connection of First Line and Second Line]

(23) The longer first line **30L** is electrically connected to the shorter second line **38S** through the one lead pin **20B**. The shorter first line **30S** is electrically connected to the longer second line **38L** through the other lead pin **20A**. This can cancel phase shift of the differential signal pair and stabilize a high-frequency signal.

(24) [Photoelectric Device]

(25) The optical module **100** has a photoelectric device **42** for converting an optical signal and an electrical signal at least from one to another. The photoelectric device **42** is mounted on the mounting substrate **44**. The mounting substrate **44** deviates in a direction closer to the first substrate **28** from the center of the first surface **12**.

(26) The mounting substrate **44** has a first electrode **46** and a second electrode **48**. The bottom surface of the photoelectric device **42** is opposed to and electrically connected to the first electrode **46**. A pad on the upper surface of the photoelectric device **42** is electrically connected to the second electrode **48** through a wire **W1**. Each of the first electrode **46** and second electrode **48** is electrically connected to a corresponding one of the pair of first lines **30S**, **30L**, through one or more wires **W2**. Thus, the photoelectric device **42** is electrically connected to the pair of first lines **30S**, **30L**.

(27) [Thermoelectric Cooler]

(28) The optical module **100** has a thermoelectric cooler **52**. The thermoelectric cooler **52** has an upper surface **54** and a lower surface **56**. The upper surface **54** and the lower surface **56** are made of an insulator such as ceramics. The lower surface **56** is fixed to the first surface **12** of the conductive block **10**. A thermally conductive adhesive may be used for fixing. The thermoelectric cooler **52** contains an unillustrated Peltier device therein for transferring heat between the upper surface **54** and the lower surface **56**. For example, the upper surface **54** is a heat absorbing surface and the lower surface **56** is a heat radiating surface, or vice versa by switching. The thermoelectric cooler **52** at the lower surface **56** is fixed to a center of the first surface **12** of the conductive block **10**. Thermoelectric cooler **52** has an electrode **50** connected to a lead pin **L1** by a wire **W3**.

(29) The thermoelectric cooler **52** has a conductive layer **58** laminated on the upper surface **54**. The conductive layer **58** is a reference potential plane (e.g., ground plane). Two or more wires **W4** connect the first surface **12** of the conductive block **10** and the conductive layer **58**. Thus, the potential of the conductive layer **58** is the same as the conductive block **10** and is stabilized. A thermistor **60** rests on and is electrically connected to the conductive layer **58** for measuring temperature. The thermistor **60** is connected to a lead pin **L2** through a wire **W5** to be supplied with a voltage.

(30) [Support Block]

(31) The optical module **100** has a support block **62**. The support block **62** is fixed to the upper surface **54** (above the conductive layer **58**) of the thermoelectric cooler **52**. The support block **62** deviates in a direction closer to the first substrate **28** from the center of the first surface **12**. The mounting substrate **44** is mounted on the support block **62**.

(32) FIG. **5** is a diagram of frequency characteristics, in a reference example, obtained by simulation using a three-dimensional electric field analysis tool. **Sdd21** indicates transmission properties of the differential signal pair. **Scd11** indicates reflective properties of the reflected and returned common-mode components. In the reference example, the lengths of the pair of lines of the differential transmission line pair are kept different, and the differential signal pair is out of phase. Therefore, differential mode noises are converted to common mode noises, apparently affecting the signal quality.

(33) FIG. **6** is a diagram of frequency characteristics, in the embodiment, obtained by simulation using the three-dimensional electric field analysis tool. In the present embodiment, as described above, the phase shift of the differential signal pair is canceled. As a result, compared with the reference example, the conversion to the common mode is suppressed, apparently improving the reflection characteristics (**Scd11**).

(34) The embodiments described above are not limited and different variations are possible. The structures explained in the embodiments may be replaced with substantially the same structures and other structures that can achieve the same effect or the same objective.

Claims

1. An optical module comprising: a conductive block having a first surface and a second surface; a pair of lead pins passing through and insulated from the conductive block, the pair of lead pins having respective first portions protruding from the first surface and respective second portions protruding from the second surface; a first substrate having a pair of first lines with different lengths, each of the pair of first lines connected to a corresponding one of the first portions, the pair of first lines being a first differential transmission line pair; a second substrate having a pair of second lines with different lengths, each of the pair of second lines connected to a corresponding one of the second portions, the pair of second lines being a second differential transmission line pair; and a photoelectric device electrically connected to the pair of first lines, the photoelectric device configured to perform one of the following functions: 1) convert an optical signal to an electrical signal or 2) convert an electrical signal to an optical signal, wherein a longer one of the pair of first lines is electrically connected to a shorter one of the pair of second lines, through a corresponding one of the pair of lead pins, a shorter one of the pair of first lines is electrically connected to a longer one of the pair of second lines, through another corresponding one of the pair of lead pins, the first substrate has a surface on which the pair of first lines are formed, and which extends along a first axis, perpendicular to the first surface, the second substrate has a surface on which the pair of second lines are formed, and which extends along a second axis, parallel to the first surface, the pair of lead pins extend along the first axis and are arranged adjacent to each other along the second axis, the pair of first lines are arranged adjacent to each other along the second axis, and the pair of second lines each have first portions extending from the pair of lead pins along

a third axis perpendicular to both the first axis and the second axis, second curved portions, and third portions extending along the second axis.

2. The optical module according to claim 1 wherein the third portions extend along the second axis in a direction from the shorter one to the longer one of the pair of first lines.
